Homework 01 - 2.2, 2.3

Due Feb 13
Uzair Hamed Mohammed

2.2: Sample Spaces and the Algebra of Sets

12. Consider the experiment of choosing coefficients for the quadratic equation $ax^2 + bx + c = 0$. Characterize the values of a, b, and c associated with the event A: Equation has complex roots.

    Sol:

    $$\text{Sample space } S : \{a, b, c \in \mathbb{R}\}$$

    $$\text{Complex roots occur when discriminant } \boxed{b^2 - 4ac < 0}$$

16. Sketch the regions in the xy-plane corresponding to $A \cup B$ and $A \cap B$ if

    $$A = \{(x, y) : 0 < x < 3, 0 < y < 3\}$$

    and

    $$B = \{(x, y) : 2 < x < 4, 2 < y < 4\}$$

    Sol:

22. Suppose that each of the twelve letters in the word

    $$T\ E\ S\ S\ E\ L\ L\ A\ T\ I\ O\ N$$

    is written on a chip. Define the events F, R and C as follows:

    F: letters in first half of alphabet

    R: letters that are repeated

    V: letters that are vowels

Which chips make up the following events?

a. $F \cap R \cap V$
b. $F^C \cap R \cap V^C$
c. $F \cap R^C \cap V$

Sol:

$$\begin{aligned} F &= \{A, E, I, L\} \\ R &= \{E, I, L, S, T\} \\ V &= \{A, E, I, O\} \end{aligned}$$

(a) $F \cap R \cap V = \boxed{\{E, I\}}$

(b) $F^C \cap R \cap V^C = \boxed{\{S, T\}}$

(c) $F \cap R^C \cap V = \boxed{\{A\}}$

38. Let A, B, C be any three events defined on a sample space S. Let $N(A), N(B), N(C), N(A \cap B), N(A \cap C), N(B \cap C), N(A \cap B \cap C)$ denote the numbers of outcomes in all the different intersections in which A, B, and C are involved. Use a Venn diagram to suggest a formula for $N(A \cup B \cup C)$.

Sol:

$$N(A \cup B \cup C) = \boxed{N(A) + N(B) + N(C) - N(A \cap B) - N(A \cap C) - N(B \cap C) + N(A \cap B \cap C)}$$

2.3: The Probability Function

10. An urn contains twenty-four chips, numbered 1 through 24. One is drawn at random. Let A be the event that the number is divisible by 2 and let B be the event that the number is divisible by 3. Find $P(A \cup B)$.

Sol:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{12}{24} + \frac{8}{24} - \frac{4}{24} \\ &= \frac{16}{24} \\ &= \boxed{\frac{2}{3}} \end{aligned}$$

12. Events A_1 and A_2 are such that $A_1 \cup A_2 = S$ and $A_1 \cap A_2 = \emptyset$. Find p_2 if $P(A_1) = p_1, P(A_2) = p_2$, and $3p_1 - p_2 = \frac{1}{2}$.

Sol:

$$\boxed{p_1 = \frac{3}{8}, p_2 = \frac{5}{8}}$$

14. Three events- A, B, C- are defined on a sample space S. Given that $P(A) = 0.2, P(B) = 0.1$, and $P(C) = 0.3$, what is the smallest possible value for $P[(A \cup B \cup C)^C]$?

<u>Sol</u>:

$$\begin{aligned} A \cap B = A \cap C = B \cap C &= \emptyset \\ &= 1 - 0.2 - 0.1 - 0.3 \\ &= \boxed{0.5} \end{aligned}$$

Homework 02 - 2.4

Due Feb 20
Uzair Hamed Mohammed

10. Suppose events A and B are such that $P(A \cap B) = 0.1$ and $P((A \cup B)^C) = 0.3$. If $P(A) = 0.2$, what does $P[(A \cap B)|(A \cup B)^C]$ equal?

    Sol:

$$P((A \cap B)|(A \cup B)^C) = \frac{P((A \cap B) \cap (A \cup B)^C)}{P((A \cap B)^C)}$$
$$(A \cap B) \cap (A \cup B)^C = \emptyset$$
$$P((A \cap B) \cap (A \cup B)^C) = 0$$
$$\boxed{P[(A \cap B)|(A \cup B)^C] = 0}$$

20. Andy, Bob, and Charley have all been serving time for grand theft auto. According to prison scuttlebutt, the warden plans to release two of the three next week. They all have identical records, so the two to be released will be chosen at random, meaning that each has a two-thirds probability of being included in the two to be set free. Andy, however, is friends with a guard who will know ahead of time which two will leave. He offers to tell Andy the name of one prisoner other than himself who will be released. Andy, however, declines the offer, believing that if he learns the name of one prisoner scheduled to be released, then his chances of being the other person set free will drop to one-half (since only two prisoners will be left at that point). Is his concern justified?

    Sol: Let A, B, and C be the events Andy, Bob, and Charley will be released respectively. Then:

$$P(A|B \cup C) = \frac{P(A \cap (B\ cupC))}{P(B \cup C)} = \frac{P((A \cap B) \cup (A \cap C))}{P(B \cup C)}$$
$$= \frac{P(\{(1,1,0)\} \cup \{(1,0,1)\})}{P(\{(0,1,1),(1,1,0),(1,0,1)\})} = \frac{\frac{2}{3}}{1} = \boxed{\frac{2}{3}}$$

22. A man has n keys on a key ring, one of which opens the door to his apartment. Having celebrated a bit too much one evening, he returns home only to find himself unable to distinguish one key from another. Resourceful, he works out a fiendishly clever plan: He will choose a key at random and try it. If it fails to open the door, he will discard it and choose at random one of the remaining $n - 1$ keys, and so on. Clearly, the probability that he gains entrance with the first key he selects is $1/n$. Show that the probability the door opens with the third key he tries is also $1/n$. (Hint: What has to happen before he even gets to the third key?)

Sol: Let K_i = event key works on the ith try:

$$\begin{aligned} P(K_3) &= P(K_1^C K_2^C K_3^C) \\ &= P(K_3|K_1^C K_2^C)P(K_2^C|K_1^C)P(K_1^C) \\ &= \boxed{(\frac{n-1}{n})(\frac{n-2}{n-1})(\frac{1}{n-2}) = \frac{1}{n}} \end{aligned}$$

30. Urn I contains three red chips and one white chip. Urn II contains two red chips and two white chips. One chip is drawn from each urn and transferred to the other urn. Then a chip is drawn from the first urn. What is the probability that the chip ultimately drawn from urn I is red?

    Sol: Let R_1 be the event a red chip was drawn from urn 1. Let R_2 be the event it was drawn from urn 2. Let R be the event it was drawn from urn 1 after the other two have been transferred:

$$\begin{aligned} P = P(R|R_1, R_2)P(R_1, R_2) + \\ P(R|R_1, R_2^C)P(R_1, R_2^C) \\ +P(R|R_1^C, R_2) + P(R|R_1^C, R_C^2)P(R_1^C, R_2^C) \\ = (\frac{3}{4})(\frac{3}{4})(\frac{2}{4}) + (\frac{2}{4})(\frac{3}{4})(\frac{2}{4}) + 1(\frac{1}{4})(\frac{2}{4}) + (\frac{3}{4})(\frac{1}{4})(\frac{2}{4}) = \boxed{0.69} \end{aligned}$$

40. Urn I contains two white chips and one red chip; urn II has one white chip and two red chips. One chip is drawn at random from urn I and transferred to urn II. Then one chip is drawn from urn II. Suppose that a red chip is selected from urn II. What is the probability that the chip transferred was white?

    Sol: Let W be the event the white chip is transferred over, and W^C be the event the red chip is transferred over. Let D be the event we draw a red chip from urn number 2:

$$\begin{aligned} P(W|D) &= \frac{P(D|W)P(W)}{P(D|W)P(W) + P(D|W^C)P(W^C)} \\ &= \frac{(\frac{2}{4})(\frac{2}{3})}{(\frac{2}{4})(\frac{2}{3}) + (\frac{1}{4})(\frac{1}{3})} \\ &= \boxed{\frac{4}{5}} \end{aligned}$$

46. Brett and Margo have each thought about murdering their rich Uncle Basil in hopes of claiming their inheritance a bit early. Hoping to take advantage of Basil's predilection for immoderate desserts, Brett has put rat poison into the cherries flambé; Margo, unaware of Brett's activities, has laced the chocolate mousse with cyanide. Given the amounts likely to be eaten, the probability of the rat poison being fatal is 0.60; the cyanide,

0.90. Based on other dinners where Basil was presented with the same dessert options, we can assume that he has a 50% chance of asking for the cherries flambé, a 40% chance of ordering the chocolate mousse, and a 10% chance of skipping dessert altogether. No sooner are the dishes cleared away than Basil drops dead. In the absence of any other evidence, who should be considered the prime suspect?

Sol: Let's define some events:

(a) F: Uncle Basil will die

(b) C: Uncle Basil chooses cherries flambé

(c) M: Uncle Basil chooses chocolate mousse

(d) N: Uncle Basil chooses no dessert

We know:

(a) $P(F|C) = 0.6$

(b) $P(F|M) = 0.9$

(c) $P(C) = 0.5$

(d) $P(M) = 0.4$

(e) $P(N) = 0.1$

So...

$$\begin{aligned} P(C|F) &= \frac{P(F|C)P(C)}{P(F|C)P(C) + P(F|M)P(M) + P(F|N)P(N)} \\ &= \frac{(0.6)(0.5)}{(0.6)(0.5) + (0.9)(0.4) + 0} \\ &= \boxed{0.45} \end{aligned}$$

and..

$$\begin{aligned} P(M|F) &= \frac{P(F|M)P(M)}{P(F|C)P(C) + P(F|M)P(M) + P(F|N)P(N)} \\ &= \frac{(0.9)(0.4)}{(0.6)(0.5) + (0.9)(0.4) + 0} \\ &= \boxed{0.55} \end{aligned}$$

Therefore, we can conclude that the chocolate mousse is most likely to be picked and Margo will be the suspect.

Homework 03 - 2.5

Due Feb 27
Uzair Hamed Mohammed

6. Three points, X_1, X_2, and X_3, are chosen at random in the interval $(0, a)$. A second set of three points, Y_1, Y_2, and Y_3, are chosen at random in the interval $(0, b)$. Let A be the event that X_2 is between X_1 and X_3. Let B be the event that $Y_1 < Y_2 < Y_3$. Find $P(A \cap B)$.

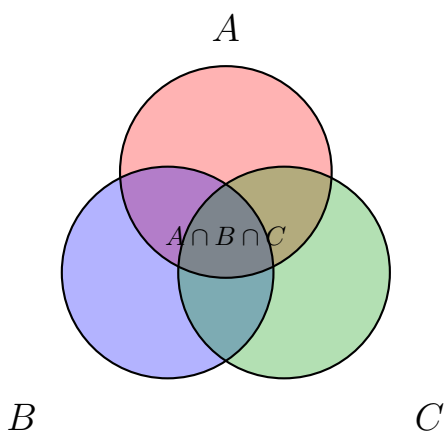
   Sol: We can expect the second random variable Y_2 to be in the middle. Given this information, as events A and B are independent, we get:

$$P(A \cap B) = P(A)P(B) = \boxed{\frac{1}{9}}$$

8. Suppose that events A, B, and C are independent.

   (a) Use a Venn diagram to find an expression for $P(A \cup B \cup C)$ that does not make use of a complement.

   Sol:



$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - [P(A)P(B) + P(A)P(C) + P(B)P(C)] + P(A)P(B)P(C)$$

   (b) Find an expression for $P(A \cup B \cup C)$ that does make use of a complement.

   Sol:

$$\begin{aligned} 1 - P((A \cup B \cup C)^C) &= 1 - P(A^C \cap B^C \cap C^C) \\ &= 1 - P(A^C)P(B^C)P(C^C) \end{aligned}$$

14. In a roll of a pair of fair dice (one red and one green), let A be the event the red die shows a 3, 4, or 5; let B be the event the green die shows a 1

or a 2; and let C be the event the dice total is 7. Show that A, B, and C are independent.

<u>Sol</u>: If we consider the probabilities of different events we will see the equities needed for independence clearly:

- $P(A) = \frac{18}{36} = \frac{1}{2}$
- $P(B) = \frac{12}{36} = \frac{1}{3}$
- $P(C) = \frac{6}{36} = \frac{1}{6}$
- $P(A \cap B) = \frac{6}{36} = \frac{1}{6} = P(A)P(B)$
- $P(A \cap C) = \frac{3}{36} = \frac{1}{12} = P(A)P(C)$
- $P(B \cap C) = \frac{2}{36} = \frac{1}{18} = P(B)P(C)$
- $P(A \cap B \cap C) = \frac{1}{36} = P(A)P(B)P(C)$

20. Players A, B, and C toss a fair coin in order. The first to throw a head wins. What are their respective chances of winning?

<u>Sol</u>: For person A,

$$\begin{aligned} P(A) &= \frac{1}{2} + (\frac{1}{2})^3 \frac{1}{2} + (\frac{1}{2}) + \dots \\ &= \frac{1}{2} \sum_{k=0}^{\infty} (\frac{1}{2})^{3k} \\ &= \frac{1}{2} (\frac{1}{1 - \frac{1}{8}}) \\ &= \boxed{\frac{4}{7}} \end{aligned}$$

For person B,

$$\begin{aligned} P(B) &= \frac{1}{2}(\frac{1}{2}) + \frac{1}{2}(\frac{1}{2})^3 \frac{1}{2} + \frac{1}{2}(\frac{1}{2})^6 \frac{1}{2} + \dots \\ &= \frac{1}{4} \sum_{k=0}^{\infty} (\frac{1}{2})^{3k} \\ &= \frac{1}{4} (\frac{1}{1 - \frac{1}{8}}) \\ &= \boxed{\frac{2}{7}} \end{aligned}$$

For person C,

$$\begin{aligned}
P(C) &= (\frac{1}{2})^2(\frac{1}{2}) + (\frac{1}{2})^2(\frac{1}{2})^3\frac{1}{2} + (\frac{1}{2})^2(\frac{1}{2})^6\frac{1}{2} + \dots \\
&= \frac{1}{8}\sum_{k=0}^{\infty}(\frac{1}{2})^{3k} \\
&= \frac{1}{8}(\frac{1}{1-\frac{1}{8}}) \\
&= \boxed{\frac{1}{7}}
\end{aligned}$$

24. Each of m urns contains three red chips and four white chips. A total of r samples with replacement are taken from each urn. What is the probability that at least one red chip is drawn from at least one urn?

    <u>Sol</u>: Since each urn has the same distribution of red and white chips,

    - $P(\text{red chip}) = \frac{3}{7}$
    - $P(\text{no red chip from r draws in a single urn}) = (1-p)^r$
    - $P(\text{no red chip from any of r draws from each of the m urns}) = (1-p)^{rm}$
    - $P(\text{at least one red chip}) = \boxed{1-(1-p)^{rm}}$

Homework 04 - 2.6, 2.7

Due Mar 6
Uzair Hamed Mohammed

2.6 Combinatorics

6. A fast-food restaurant offers customers a choice of eight toppings that can be added to a hamburger. How many different hamburgers can be ordered?

   Sol: $\boxed{2^8 = 256}$ different hamburgers can be ordered since each topping can be added for two choices of each topping.

16. Monica's vacation plans require that she fly from Nashville to Chicago to Seattle to Anchorage. According to her travel agent, there are three available flights from Nashville to Chicago, five from Chicago to Seattle, and two from Seattle to Anchorage. Assume that the numbers of options she has for return flights are the same. How many round-trip itineraries can she schedule?

    Sol: On her departing trip, she can schedule $3 \times 5 \times 2 = 30$ itineraries. On her way back, she can schedule the same number. So, in total, she can schedule $\boxed{30^2}$ different round-trip itineraries.

18. How many ways can a set of four tires be put on a car if all the tires are interchangeable? How many ways are possible if two of the four are snow tires?

    Sol: ${}_2P_0 = 1$ if all the tires are interchangeable, and we can select the location of the first snow tire in four ways and the location of the second snow tire in three ways to make a total of 12 ways in the second part.

26. A new horror movie, Friday the 13th, Part X, will star Jason's great-grandson (also named Jason) as a psychotic trying to dispatch (as gruesomely as possible) eight camp counselors, four men and four women. (a) How many scenarios (i.e., victim orders) can the screenwriters devise, assuming they want Jason to do away with all the men before going after any of the women? (b) How many scripts are possible if the only restriction imposed on Jason is that he save Muffy for last?

    (a) Sol: There are 4! ways to order the four men and 4! ways to order the four women, meaning there are $\boxed{4! \times 4! = 576}$ scenarios where the men are dispatched before the women.

    (b) Sol: Since Muffy is fixed in the last position, there are 7! ways to arrange the remaining seven camp counselors, resulting in $\boxed{7! = 5040}$ possible scripts

38. How many ways can the letters in the word

$$SLUMGULLION$$

be arranged so that the three L's precede all the other consonants?

<u>Sol</u>: There are $\binom{11}{4} = 330$ ways to choose the positions for the vowels and $\frac{4!}{2!} = 12$ ways to arrange the set $\{U, U, I, O\}$. Since the three Ls must precede the other four consonants, they must occupy the first three consonant positions in the sequence, leaving $4! = 24$ ways to arrange the remaining consonants (S, M, G, N), resulting in a total of $\boxed{330 \times 12 \times 24 = 95,040}$ arrangements.

58. Prove that

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$$

<u>Sol</u>: Expanding the right-hand side with factorials gives $\frac{n!}{k!(n-k)!} + \frac{n!}{(k-1)!(n-k+1)!}$. By using a common denominator of $k!(n-k+1)!$, we sum the numerators to get $n!(n-k+1+k) = n!(n+1)$, which simplifies to $\frac{(n+1)!}{k!(n+1-k)!} = \boxed{\binom{n+1}{k}}$.

2.7 Combinatorial Probability

6. A committee of fifty politicians is to be chosen from among one hundered US senators. If the selection is done at random, what is the probability that each state will be represented?

<u>Sol</u>: We can choose 50 politicians in $\binom{100}{50}$ ways. For each state, we can choose one representative in $\binom{2}{1} = 2$ ways. In the end, we get probability:

$$\boxed{P = \frac{2^{50}}{\binom{100}{50}}}$$

12. If the letters in the phrase

$$A\ ROLLING\ STONE\ GATHERS\ NO\ MOSS$$

are arranged at random, what are the chances that not all the Ss will be adjacent?

First, we have the following amount of unique permutations of all of the 26 letters:

$$\frac{26!}{4!4!3!2!2!2!2!2!2!}$$

Among these permutations, one is a subset contains all of the Ss adjacent. We can order all of these in

$$\boxed{\frac{(26-4+1)!}{1!4!3!2!2!2!2!2!2!}}$$

different ways.

20. Five cards are dealt from a standard 52-card deck. What is the probability that the sum of the faces on the five cards is 48 or more?

    Sol: In a standard 52-card deck, if we assign the ranks Ace through King the values (1, 2, ..., 13), the total number of ways to be dealt five cards is $\binom{52}{5} = 2,598,960$. To find the number of hands where the sum of the faces is 48 or more, we sum the combinations of ranks that satisfy this condition (considering there are 4 of each rank). There are (158,472) such combinations, making the probability:

$$P(\text{Sum} \geq 48) = \frac{158,472}{2,598,960} \approx \boxed{0.061}$$

Homework 05 - 3.2, 3.3

Due Mar 13
Uzair Hamed Mohammed

3.2 Binomial and Hypergeometric Probabilities

10. The gunner on a small assault boat fires six missiles at an attacking plane. Each has a 20% chance of being on-target. If two or more of the shells find their mark, the plane will crash. At the same time, the pilot of the plane fires ten air-to-surface rockets, each of which has a 0.05 chance of critically disabling the boat. Would you rather be on the plane or the boat?

    Sol:

    Plane Survival Probability:

    $$P(X \geq 2) = 1 - \left[\binom{6}{0}(0.2)^0(0.8)^6 + \binom{6}{1}(0.2)^1(0.8)^5\right]$$

    $$P(X \geq 2) = 1 - [0.262144 + 0.393216] = 0.34464$$

    $$P(\text{Survival}_{\text{Plane}}) = 1 - 0.34464 = 0.65536$$

    Boat Survival Probability:

    $$P(Y \geq 1) = 1 - \binom{10}{0}(0.05)^0(0.95)^{10}$$

    $$P(Y \geq 1) = 1 - 0.598737 = 0.401263$$

    $$P(\text{Survival}_{\text{Boat}}) = 1 - 0.401263 = 0.598737$$

    $$P(\text{Survival}_{\text{Plane}}) > P(\text{Survival}_{\text{Boat}})$$

    $$0.65536 > 0.598737$$

    $$\boxed{\text{Preference: Plane}}$$

12. Experience has shown that only $\frac{1}{3}$ of all patients having a certain disease will recover if given the standard treatment. A new drug is to be tested on a group of twelve volunteers. If the FDA requires that at least seven of these patients recover before it will license the new drug, what is the probability that the treatment will be discredited even if it has the potential to increase an individual's recovery rate to $\frac{1}{2}$?

    Sol:

Parameters:

$n = 12$

$p = 0.5$

The drug is discredited if fewer than 7 patients recover:

$$P(X < 7) = P(X \leq 6) = \sum_{k=0}^{6} \binom{12}{k} (0.5)^k (0.5)^{12-k}$$

$$P(X \leq 6) = \frac{1}{2^{12}} \sum_{k=0}^{6} \binom{12}{k}$$

$$\sum_{k=0}^{6} \binom{12}{k} = \binom{12}{0} + \binom{12}{1} + \binom{12}{2} + \binom{12}{3} + \binom{12}{4} + \binom{12}{5} + \binom{12}{6}$$

$$\sum_{k=0}^{6} \binom{12}{k} = 1 + 12 + 66 + 220 + 495 + 792 + 924 = 2510$$

$$P(X \leq 6) = \frac{2510}{4096} \approx \boxed{0.6128}$$

14. The captain of a Navy gunboat orders a volley of twenty-five missiles to be fired at random along a five-hundred-foot stretch of shoreline that he hopes to establish as a beachhead. Dug into the beach is a thirty-foot-long bunker serving as the enemy's first line of defense. The captain has reason to believe that the bunker will be destroyed if at least three of the missiles are on-target. What is the probability of that happening?

    Sol:

Parameters:

$n = 25$

$p = \dfrac{30}{500} = 0.06$

Bunker is destroyed if $X \geq 3$:

$$P(X \geq 3) = 1 - \sum_{k=0}^{2} \binom{25}{k} (0.06)^k (0.94)^{25-k}$$

$$P(X = 0) = \binom{25}{0} (0.06)^0 (0.94)^{25} \approx 0.2129$$

$$P(X = 1) = \binom{25}{1} (0.06)^1 (0.94)^{24} \approx 0.3398$$

$$P(X = 2) = \binom{25}{2} (0.06)^2 (0.94)^{23} \approx 0.2602$$

$$P(X \geq 3) = 1 - (0.2129 + 0.3398 + 0.2602)$$

$$P(X \geq 3) = 1 - 0.8129 \approx \boxed{0.1871}$$

16. Listed in the following table is the length distribution of World Series competition for the sixty-four series from 1950 to 2014 (there was no series in 1994). Assuming that each World Series game is an independent event and that the probability of either team's winning any particular contest is 0.5, find the probability of each series length. How well does the model fit the data? (Compute the "expected" frequencies, that is, multiply the probability of a given-length series times 64).

    Sol:

Theoretical Probabilities $P(k)$ for k games:

$P(k) = 2 \times \binom{k-1}{3}(0.5)^k$

$P(4) = 2 \times \binom{3}{3}(0.5)^4 = 2(1)(0.0625) = 0.125$

$P(5) = 2 \times \binom{4}{3}(0.5)^5 = 2(4)(0.03125) = 0.25$

$P(6) = 2 \times \binom{5}{3}(0.5)^6 = 2(10)(0.015625) = 0.3125$

$P(7) = 2 \times \binom{6}{3}(0.5)^7 = 2(20)(0.0078125) = 0.3125$

Expected Frequencies ($E_k = P(k) \times 64$) :

$E_4 = 0.125 \times 64 = 8.0$

$E_5 = 0.25 \times 64 = 16.0$

$E_6 = 0.3125 \times 64 = 20.0$

$E_7 = 0.3125 \times 64 = 20.0$

Comparison with Observed Data (1950–2014):

k	Observed (O_k)	Expected (E_k)	$\lvert O_k - E_k \rvert$
4	12	8.0	4.0
5	11	16.0	5.0
6	14	20.0	6.0
7	27	20.0	7.0

Model Fit: The model fits poorly. It significantly underestimates the frequency of four-game and seven-game series.

20. A corporate board contains twelve members. The board decides to create a five-person Committee to Hide Corporation Debt. Suppose four members of the board are accountants. What is the probability that the Committee will contain two accountants and three non-accountants?

    Sol:

Total members: $N = 12$

Accountants: $k = 4$

Non-accountants: $N - k = 8$

Committee size: $n = 5$

$$P(X = 2) = \frac{\binom{4}{2}\binom{8}{3}}{\binom{12}{5}}$$

$$\binom{4}{2} = \frac{4 \times 3}{2 \times 1} = 6$$

$$\binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$$

$$\binom{12}{5} = \frac{12 \times 11 \times 10 \times 9 \times 8}{5 \times 4 \times 3 \times 2 \times 1} = 792$$

$$P(X = 2) = \frac{6 \times 56}{792} = \frac{336}{792} \approx \boxed{0.4242}$$

22. A city has 4050 children under the age of ten, including 514 who have not been vaccinated for measles. Sixty-five of the city's children are enrolled in the ABC Day Care Center. Suppose the municipal health department sends a doctor and a nurse to ABC to immunize any child who has not already been vaccinated. Find a formula for the probability that exactly k of the children at ABC have not been vaccinated.

    Sol:

    Total children (N) = 4050

    Unvaccinated (M) = 514

    Vaccinated (N - M) = 3536

    Sample size (n) = 65

    Hypergeometric Distribution Formula:

    $$\boxed{P(X = k) = \frac{\binom{514}{k}\binom{3536}{65-k}}{\binom{4050}{65}}, \quad k = 0, 1, \ldots, 65}$$

27. A display case contains thirty-five gems, of which ten are real diamonds and twenty-five are fake diamonds. A burglar removes four gems at random, one at a time and without replacement. What is the probability that the last gem she steals is the second real diamond in the set of four?

    Sol:

    The burglar has to first draw a single diamond in the first set of three

gems removed. Then, the burglar has to draw a second diamond on his fourth draw. We can calculate the probability of this to be $\boxed{0.1289152}$

28. Consider an urn with r red balls and w white balls, where $r + w = N$. Draw n balls in order without replacement. Show that the probability of k red balls is hypergeometric.

    Sol:

    Total balls: $N = r + w$

    Number of draws: n

    Number of red balls desired: k

    Total number of ways to choose n balls from N (ignoring order):

    $$\binom{N}{n}$$

    Number of ways to choose k red balls from r:

    $$\binom{r}{k}$$

    Number of ways to choose the remaining $(n-k)$ white balls from w:

    $$\binom{w}{n-k}$$

    By the Multiplication Principle, the number of successful outcomes is:

    $$\binom{r}{k}\binom{w}{n-k}$$

    The probability of k red balls is the ratio of favorable to total outcomes:

    $$P(X = k) = \frac{\binom{r}{k}\binom{w}{n-k}}{\binom{N}{n}}, \quad \max(0, n-w) \leq k \leq \min(n, r)$$

    This is the PMF of the Hypergeometric Distribution.

30. Show that the ratio of two successive hypergeometric probability terms satisfies the following equation,

$$\frac{\binom{r}{k+1}\binom{w}{n-k-1}}{\binom{N}{n}} \div \frac{\binom{r}{k}\binom{w}{n-k}}{\binom{N}{n}} = \frac{n-k}{k+1} \cdot \frac{r-k}{w-n+k+1}$$

Sol:

Let $P(k) = \frac{\binom{r}{k}\binom{w}{n-k}}{\binom{N}{n}}$ and $P(k+1) = \frac{\binom{r}{k+1}\binom{w}{n-k-1}}{\binom{N}{n}}$.

$$\frac{P(k+1)}{P(k)} = \frac{\binom{r}{k+1}\binom{w}{n-k-1}}{\binom{N}{n}} \cdot \frac{\binom{N}{n}}{\binom{r}{k}\binom{w}{n-k}} = \frac{\binom{r}{k+1}}{\binom{r}{k}} \cdot \frac{\binom{w}{n-k-1}}{\binom{w}{n-k}}$$

Expanding the first term:

$$\frac{\binom{r}{k+1}}{\binom{r}{k}} = \frac{\frac{r!}{(k+1)!(r-k-1)!}}{\frac{r!}{k!(r-k)!}} = \frac{k!(r-k)!}{(k+1)!(r-k-1)!} = \frac{k!}{(k+1)k!} \cdot \frac{(r-k)(r-k-1)!}{(r-k-1)!} = \frac{r-k}{k+1}$$

Expanding the second term:

$$\frac{\binom{w}{n-k-1}}{\binom{w}{n-k}} = \frac{\frac{w!}{(n-k-1)!(w-(n-k-1))!}}{\frac{w!}{(n-k)!(w-(n-k))!}} = \frac{(n-k)!(w-n+k)!}{(n-k-1)!(w-n+k+1)!}$$

$$\frac{\binom{w}{n-k-1}}{\binom{w}{n-k}} = \frac{(n-k)(n-k-1)!}{(n-k-1)!} \cdot \frac{(w-n+k)!}{(w-n+k+1)(w-n+k)!} = \frac{n-k}{w-n+k+1}$$

Multiplying the simplified terms:

$$\frac{P(k+1)}{P(k)} = \frac{r-k}{k+1} \cdot \frac{n-k}{w-n+k+1} = \frac{n-k}{k+1} \cdot \frac{r-k}{w-n+k+1}$$

36. Sixteen students: five freshmen, four sophomores, four juniors, and three seniors—have applied for membership in their school's Communications Board, a group that oversees the college's newspaper, literary magazine, and radio show. Eight positions are open. If the selection is done at random, what is the probability that each class gets two representatives? (Hint: Use the generalized hypergeometric model asked for in Question 3.2.35.)

    Sol:
    Using question 3.2.35 as reference, we would get

$$\frac{\binom{5}{2}\binom{4}{2}\binom{4}{2}\binom{3}{2}}{\binom{16}{8}} \approx \boxed{0.0839}$$

3.3 Discrete Random Variables

2. An urn contains five balls numbered 1 to 5. Two balls are drawn simultaneously with replacement.

    (a) Let X be the larger of the two numbers drawn. Find $p_X(k)$.

Sol:

There are $5 \times 5 = 25$ total possible outcomes (i, j). $X = \max(i, j)$.

The event $X \leq k$ occurs if both balls are $\leq k$.

$$P(X \leq k) = \frac{k^2}{25}$$

$$p_X(k) = P(X \leq k) - P(X \leq k-1)$$

$$p_X(k) = \frac{k^2 - (k-1)^2}{25} = \frac{2k-1}{25}, \quad k = 1, 2, 3, 4, 5$$

Distribution Table for X :

k	1	2	3	4	5
$p_X(k)$	$\frac{1}{25}$	$\frac{3}{25}$	$\frac{5}{25}$	$\frac{7}{25}$	$\frac{9}{25}$

(b) Let V be the sum of the two numbers drawn. Find $p_V(k)$.
Sol:
$V = i + j$, where $i, j \in \{1, 2, 3, 4, 5\}$. The possible sums range from $1 + 1 = 2$ to $5 + 5 = 10$.

Outcomes for each sum k :

$k = 2 : \{(1, 1)\} \implies 1$

$k = 3 : \{(1, 2), (2, 1)\} \implies 2$

$k = 4 : \{(1, 3), (2, 2), (3, 1)\} \implies 3$

$k = 5 : \{(1, 4), (2, 3), (3, 2), (4, 1)\} \implies 4$

$k = 6 : \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\} \implies 5$

$k = 7 : \{(2, 5), (3, 4), (4, 3), (5, 2)\} \implies 4$

$k = 8 : \{(3, 5), (4, 4), (5, 3)\} \implies 3$

$k = 9 : \{(4, 5), (5, 4)\} \implies 2$

$k = 10 : \{(5, 5)\} \implies 1$

Distribution Table for V :

k	2	3	4	5	6	7	8	9	10
$p_V(k)$	$\frac{1}{25}$	$\frac{2}{25}$	$\frac{3}{25}$	$\frac{4}{25}$	$\frac{5}{25}$	$\frac{4}{25}$	$\frac{3}{25}$	$\frac{2}{25}$	$\frac{1}{25}$

8. Suppose a particle moves along the x-axis beginning at 0. It moves one integer step to the left with some probability. It is twice as likely to move to the right as to the left. What is the pdf of its position after four steps?

Sol:

Let p be the probability of moving left (-1) and q be the probability of moving right $(+1)$.
$q = 2p$

$p + q = 1 \implies p + 2p = 1 \implies p = \dfrac{1}{3}, \quad q = \dfrac{2}{3}$

Let X be the position after $n = 4$ steps. Let k be the number of steps to the right.
The number of steps to the left is $4 - k$.
Position $X = k - (4 - k) = 2k - 4$.

Possible values for $k \in \{0, 1, 2, 3, 4\}$:
$k = 0 \implies X = -4$
$k = 1 \implies X = -2$
$k = 2 \implies X = 0$
$k = 3 \implies X = 2$
$k = 4 \implies X = 4$

Probability mass function $p_X(x) = \binom{4}{k} q^k p^{4-k}$:

$$p_X(-4) = \binom{4}{0}\left(\frac{2}{3}\right)^0\left(\frac{1}{3}\right)^4 = 1 \cdot 1 \cdot \frac{1}{81} = \frac{1}{81}$$

$$p_X(-2) = \binom{4}{1}\left(\frac{2}{3}\right)^1\left(\frac{1}{3}\right)^3 = 4 \cdot \frac{2}{3} \cdot \frac{1}{27} = \frac{8}{81}$$

$$p_X(0) = \binom{4}{2}\left(\frac{2}{3}\right)^2\left(\frac{1}{3}\right)^2 = 6 \cdot \frac{4}{9} \cdot \frac{1}{9} = \frac{24}{81}$$

$$p_X(2) = \binom{4}{3}\left(\frac{2}{3}\right)^3\left(\frac{1}{3}\right)^1 = 4 \cdot \frac{8}{27} \cdot \frac{1}{3} = \frac{32}{81}$$

$$p_X(4) = \binom{4}{4}\left(\frac{2}{3}\right)^4\left(\frac{1}{3}\right)^0 = 1 \cdot \frac{16}{81} \cdot 1 = \frac{16}{81}$$

PDF for X :

x	-4	-2	0	2	4
$p_X(x)$	$\frac{1}{81}$	$\frac{8}{81}$	$\frac{24}{81}$	$\frac{32}{81}$	$\frac{16}{81}$

14. At the points $x = 0, 1, \ldots, 6$, the cdf for the discrete random variable X has the value $F_X = x(x+1)/42$. Find the pdf for X.

<u>Sol</u>:

The pdf $p_X(k)$ is related to the cdf $F_X(k)$ by:
$p_X(k) = F_X(k) - F_X(k-1)$

Given $F_X(x) = \dfrac{x(x+1)}{42}$ for $x = 0, 1, \ldots, 6$:

$$p_X(k) = \frac{k(k+1)}{42} - \frac{(k-1)((k-1)+1)}{42}$$
$$p_X(k) = \frac{k^2 + k - (k-1)k}{42}$$
$$p_X(k) = \frac{k^2 + k - (k^2 - k)}{42}$$
$$p_X(k) = \frac{2k}{42} = \frac{k}{21}, \quad k = 1, 2, 3, 4, 5, 6$$

Distribution Table for X :

k	1	2	3	4	5	6
$p_X(k)$	$\frac{1}{21}$	$\frac{2}{21}$	$\frac{3}{21}$	$\frac{4}{21}$	$\frac{5}{21}$	$\frac{6}{21}$

Homework 06 - 3.4

Due Mar 20
Uzair Hamed Mohammed

6. Let n be a positive integer. Show that $f_Y(y) = (n+2)(n+1)y^n(1-y), 0 \leq y \leq 1$, is a pdf.

   Sol:

$$n \in \mathbb{Z}^+, y \in [0,1] \implies (n+2)(n+1) > 0, y^n \geq 0, (1-y) \geq 0$$

$$\begin{aligned}
\int_0^1 (n+2)(n+1)y^n(1-y)dy &= (n+2)(n+1)\int_0^1 (y^n - y^{n+1})dy \\
&= (n+2)(n+1)\left[\frac{y^{n+1}}{n+1} - \frac{y^{n+2}}{n+2}\right]_0^1 \\
&= (n+2)(n+1)\left(\frac{1}{n+1} - \frac{1}{n+2}\right) \\
&= (n+2)(n+1)\left[\frac{(n+2)-(n+1)}{(n+1)(n+2)}\right] \\
&= \frac{(n+2)(n+1)}{(n+1)(n+2)} \\
&= 1
\end{aligned}$$

9. If the pdf for Y is

$$f_Y(y) = \begin{cases} 0, & |y| > 1 \\ 1 - |y|, & |y| \leq 1 \end{cases}$$

   find and graph $F_Y(y)$.

   Sol:

$$F_Y(y) = \int_{-\infty}^{y} f_Y(t)dt$$

   For $y < -1$:

$$F_Y(y) = 0$$

   For $-1 \leq y < 0$:

$$F_Y(y) = \int_{-1}^{y} (1+t)dt = \left[t + \frac{t^2}{2}\right]_{-1}^{y} = (y + \frac{y^2}{2}) - (-1 + \frac{1}{2}) = \frac{1}{2}(y+1)^2$$

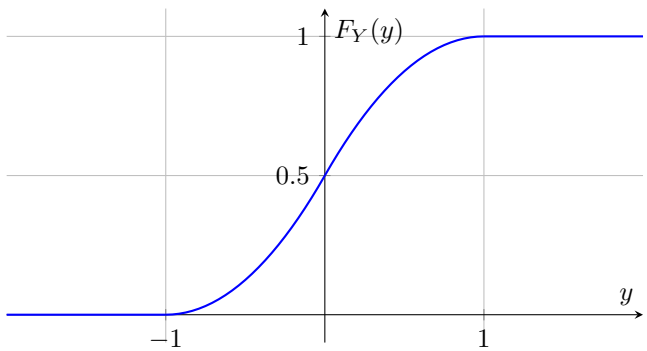
For $0 \leq y \leq 1$:

$$F_Y(y) = F_Y(0) + \int_0^y (1-t)dt = \frac{1}{2} + \left[t - \frac{t^2}{2}\right]_0^y = \frac{1}{2} + y - \frac{y^2}{2} = 1 - \frac{1}{2}(1-y)^2$$

For $y > 1$:

$$F_Y(y) = 1$$

$$F_Y(y) = \begin{cases} 0, & y < -1 \\ \frac{1}{2}(y+1)^2, & -1 \leq y < 0 \\ 1 - \frac{1}{2}(1-y)^2, & 0 \leq y \leq 1 \\ 1, & y > 1 \end{cases}$$



14. In a certain country, the distribution of a family's disposable income, Y, is described by the pdf $f_Y(y) = ye^{-y}, y \geq 0$. Find $F_Y(y)$.

<u>Sol</u>:

$$F_Y(y) = \int_0^y f_Y(t)dt = \int_0^y te^{-t}dt$$

Integration by parts: Let $u = t, dv = e^{-t}dt \implies du = dt, v = -e^{-t}$

$$\int te^{-t}dt = -te^{-t} - \int -e^{-t}dt = -te^{-t} - e^{-t} = -(t+1)e^{-t}$$

Evaluating the definite integral:

$$F_Y(y) = \left[-(t+1)e^{-t}\right]_0^y$$

$$F_Y(y) = -(y+1)e^{-y} - (-(0+1)e^0)$$

$$F_Y(y) = 1 - (y+1)e^{-y}, \quad y \geq 0$$

For $y < 0$, $F_Y(y) = 0$.

$$F_Y(y) = \begin{cases} 1 - (y+1)e^{-y}, & y \geq 0 \\ 0, & y < 0 \end{cases}$$

Homework 08 - 3.7

Due Apr 17
Uzair Hamed Mohammed

12. A point is chosen at random from the interior of a circle whose equation is $x^2 + y^2 \leq 4$. Let the random variables X and Y denote the x- and y-coordinates of the sampled point. Find $f_{X,Y}(x, y)$.

    Sol:

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{4\pi}, & x^2 + y^2 \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

14. Suppose that five independent observations are drawn from the continuous pdf $f_T(t) = 2t, 0 \leq t \leq 1$. Let X denote the number of t's that fall in the interval $0 \leq t < \frac{1}{3}$ and let Y denote the number of t's that fall in the interval $\frac{1}{3} \leq t < \frac{2}{3}$. Find $p_{X,Y}(1, 2)$.

    Sol:

$$p_{X,Y}(1,2) = \frac{5!}{1!\,2!\,2!}\left(\frac{1}{9}\right)^1\left(\frac{1}{3}\right)^2\left(\frac{5}{9}\right)^2 = 30 \cdot \frac{1}{9} \cdot \frac{1}{9} \cdot \frac{25}{81} = \frac{250}{2187}$$

22. Find $f_Y(y)$ if $f_{X,Y}(x, y) = 2e^{-x}e^{-y}$ for x and y defined over the shaded region pictured.

    Sol:

$$f_Y(y) = \begin{cases} 2e^{-y} - 2e^{-2y}, & y > 0 \\ 0, & \text{otherwise} \end{cases}$$

30. Find the joint pdf associated with two random variables X and Y whose joint cdf is

$$F_{X,Y}(x,y) = (1 - e^{-\lambda y})(1 - e^{-\lambda x}), x > 0, y > 0$$

    Sol:

$$f_{X,Y}(x,y) = \begin{cases} \lambda^2 e^{-\lambda(x+y)}, & x > 0,\ y > 0 \\ 0, & \text{otherwise} \end{cases}$$

36. Suppose that the random variables X, Y, and Z have the multivariate pdf

$$f_{X,Y,Z}(x,y,z) = (x + y)e^{-z}$$

    for $0 < x < 1, 0 < y < 1$, and $z > 0$. Find:

    (a) $f_{X,Y}(x, y)$

(b) $f_{Y,Z}(y,z)$

(c) $f_Z(z)$

<u>Sol</u>:

$$\begin{aligned}
&\text{(a)} \quad f_{X,Y}(x,y) = x+y, \quad 0<x<1,\ 0<y<1 \\
&\text{(b)} \quad f_{Y,Z}(y,z) = \left(y+\frac{1}{2}\right)e^{-z}, \quad 0<y<1,\ z>0 \\
&\text{(c)} \quad f_Z(z) = e^{-z}, \quad z>0
\end{aligned}$$

42. Are the random variables X and Y independent if $f_{X,Y}(x,y) = \frac{2}{3}(x+2y), 0 \le x \le 1, 0 \le y \le 1$?

<u>Sol</u>:

No, since $f_{X,Y}(x,y) \neq f_X(x)f_Y(y)$

52. A random sample size of $n = 2k$ is taken from a uniform pdf defined over the unit interval. Calculate $P(X_1 < \frac{1}{2}, X_2 > \frac{1}{2}, X_3 < \frac{1}{2}, X_4 > \frac{1}{2}, \ldots X_{2k} > \frac{1}{2})$

<u>Sol</u>:

$$P = \left(\frac{1}{2}\right)^{2k} = \frac{1}{4^k}$$

Homework 09 - 3.8, 3.9

Due Apr 24
Uzair Hamed Mohammed

3.8 Transforming and Combining Random Variables

2. Let $f_Y(y) = \frac{3}{14}(1+y^2), 0 \leq y \leq 2$. Define the random variable W by $W = 3Y + 2$. Find $f_W(w)$. Be sure to specify the values of w for which $f_W(w) \neq 0$.

   Sol:

$$\begin{aligned}
&\text{Given } f_Y(y) = \frac{3}{14}(1+y^2),\ 0 \leq y \leq 2. \\
&W = 3Y + 2 \implies Y = \frac{W-2}{3}, \quad \frac{dy}{dw} = \frac{1}{3}. \\
&f_W(w) = f_Y\left(\frac{w-2}{3}\right)\left|\frac{dy}{dw}\right| \\
&= \frac{3}{14}\left(1 + \left(\frac{w-2}{3}\right)^2\right) \cdot \frac{1}{3} \\
&= \frac{1}{14}\left(1 + \frac{(w-2)^2}{9}\right) \\
&= \frac{1}{14} \cdot \frac{9 + (w-2)^2}{9} \\
&= \frac{1}{126}\big(9 + (w-2)^2\big), \quad 2 \leq w \leq 8.
\end{aligned}$$

6. If a random variable V is independent of two independent random variables X and Y, prove that B is independent of X + Y.

   Sol:

$$\text{Given } V \perp X,\ V \perp Y,\ X \perp Y. \text{ Show } V \perp (X+Y).$$

$$\forall u, v:\ P(V \leq v, X+Y \leq u) = \int_{-\infty}^{\infty} P(X+Y \leq u \mid V = t) f_V(t)dt.$$

   Since $V \perp (X,Y)$, (X,Y) independent of V, so $P(X+Y \leq u \mid V = t) = P(X+Y \leq u)$.

$$\therefore P(V \leq v, X+Y \leq u) = P(X+Y \leq u)P(V \leq v).$$

   Thus V independent of $X+Y$. (The conclusion holds for $B = V$.)

8. Let Y be a uniform random variable over the interval $[0,1]$. Find the pdf of $W = Y^2$.

Sol:

$$\begin{aligned}
&\text{Let } Y \sim U(0,1),\ F_Y(y) = y,\ f_Y(y) = 1.\\
&W = Y^2,\ w \in (0,1).\\
&F_W(w) = P(W \le w) = P(Y^2 \le w) = P(Y \le \sqrt{w}) = \sqrt{w}.\\
&f_W(w) = \frac{d}{dw} F_W(w) = \frac{1}{2\sqrt{w}}, \quad 0 < w < 1.
\end{aligned}$$

3.9 Further Properties of the Mean and Variance

4. Marksmanship competition at a certain level requires each contestant to take ten shots with each of two different handguns. Final scores are computed by taking a weighted average of four times the number of bull's-eyes made with the first gun plus six times the number gotten with the second. If Cathie has a 30% chance of hitting the bull's-eye with each shot from the first gun and a 40% chance with each shot from the second gun, what is her expected score?

   Sol:

$$E[X] = 10 \cdot 0.3 = 3, \quad E[Y] = 10 \cdot 0.4 = 4$$
$$E[\text{Score}] = 4E[X] + 6E[Y] = 4 \cdot 3 + 6 \cdot 4 = 12 + 24 = 36$$

10. Suppose that X and Y are both uniformly distributed over the interval $[0,1]$. Calculate the expected value of the square of the distance of the random point (X,Y) from the origin; that is, find $E(X^2 + Y^2)$.

    Sol:

$$\begin{aligned}
&X, Y \sim U(0,1),\ \text{independent.}\\
&E(X^2) = \int_0^1 x^2\,dx = \frac{1}{3}, \quad E(Y^2) = \frac{1}{3}.\\
&E(X^2 + Y^2) = E(X^2) + E(Y^2) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}.
\end{aligned}$$

14. Show that

$$\text{Cov}(aX + b, cY + d) = ac\text{Cov}(X, Y)$$

    for any constants a, b, c, and d.

    Sol:

$$
\begin{aligned}
\mathrm{Cov}(aX+b, cY+d) &= E\big[((aX+b)-E[aX+b])((cY+d)-E[cY+d])\big] \\
&= E\big[(aX+b-aE[X]-b)(cY+d-cE[Y]-d)\big] \\
&= E\big[a(X-E[X])\cdot c(Y-E[Y])\big] \\
&= ac\,E\big[(X-E[X])(Y-E[Y])\big] \\
&= ac\,\mathrm{Cov}(X,Y).
\end{aligned}
$$

20. Let X be binomial random variable based on n trials and a success probability of p_X; let Y be an independent binomial random variable based on m trials and a success probability of p_Y. Find $E(W)$ and $\mathrm{Var}(W)$, where $W = 4X + 6Y$.

    <u>Sol</u>:

$$
\begin{aligned}
E(W) &= 4np_X + 6mp_Y, \\
\mathrm{Var}(W) &= 16np_X(1-p_X) + 36mp_Y(1-p_Y).
\end{aligned}
$$

Homework 10 - 3.10, 3.11

Due May 1
Uzair Hamed Mohammed

3.10 Order Statistics

4. A random sample of size 5 is drawn from the pdf $f_Y(y) = 2y, 0 \leq y \leq 1$. Calculate $P(Y_1' < 0.6 < Y_5')$.

   Sol:

$$F_Y(y) = y^2, \quad 0 \leq y \leq 1.$$

$$P(Y_1' < 0.6 < Y_5') = 1 - [F(0.6)]^5 - [1 - F(0.6)]^5 = 1 - (0.36)^5 - (0.64)^5.$$

$$0.36 = \frac{9}{25}, \quad 0.64 = \frac{16}{25}.$$

$$1 - \left(\frac{9}{25}\right)^5 - \left(\frac{16}{25}\right)^5 = 1 - \frac{59049 + 1048576}{9765625} = 1 - \frac{1107625}{9765625} = \frac{8658000}{9765625} = \frac{69264}{78125}.$$

$$\boxed{\frac{69264}{78125}}$$

10. Suppose that n observations are chosen at random from a continuous pdf $f_Y(y)$. What is the probability that the last observation recorded will be the smallest number in the entire sample?

    Sol:

$$\boxed{\frac{1}{n}}$$

12. Consider a system containing n components, where the lifetimes of the components are independent random variables and each has pdf $f_Y(y) = \lambda e^{-\lambda y}, y > 0$. Show that the average time elapsing before the first component failure occurs is $1/n\lambda$.

    Sol:

    Let $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} \text{Exp}(\lambda)$. The time until the first failure is $Y_{(1)} = \min(Y_1, \ldots, Y_n)$. For i.i.d. exponentials, $Y_{(1)} \sim \text{Exp}(n\lambda)$. Hence $\mathbb{E}[Y_{(1)}] = \frac{1}{n\lambda}$. $\square$

3.11 Conditional Densities

4. Five cards are dealt from a standard poker deck. Let X be the number of aces received, and Y the number of kinds. Compute $P(X = 2|Y = 2)$.

   Sol:

$$P(X = 2 \mid Y = 2) = \frac{\binom{4}{2}\binom{12}{1}\binom{4}{3}}{\binom{13}{2}\left[\binom{4}{1}\binom{4}{4} + \binom{2}{1}\binom{4}{3}\binom{4}{2}\right]} = \boxed{\frac{6}{91}}$$

8. In Question 3.11.7, define the random variable W to be the "majority" of x, y, and z. For example, $W(2, 2, 1) = 2$ and $W(1, 1, 1) = 1$. Find the pdf of $W|_x$. For reference, Question 3.11.7 goes:

   Suppose X, Y, and Z have a trivariate distribution described by the joint pdf

$$p_{X,Y,Z}(x, y, z) = \frac{xy + xz + yz}{54}$$

   where x, y, and z can be 1 or 2. Tabulate the joint conditional pdf of X and Y given each of the two values of z.

   Sol:

   Givem joint pmf $p(x, y, z) = \frac{xy+xz+yz}{54}$ for $x, y, z \in \{1, 2\}$.

   Compute all probabilities:

(x, y, z)	p
$(1, 1, 1)$	$3/54$
$(1, 1, 2)$	$5/54$
$(1, 2, 1)$	$5/54$
$(1, 2, 2)$	$8/54$
$(2, 1, 1)$	$5/54$
$(2, 1, 2)$	$8/54$
$(2, 2, 1)$	$8/54$
$(2, 2, 2)$	$12/54$

   Marginals: $P(X = 1) = \frac{21}{54} = \frac{7}{18}$, $P(X = 2) = \frac{33}{54} = \frac{11}{18}$.

   For $X = 1$: - $W = 1$ when $(1, 1, 1)$, $(1, 1, 2)$, $(1, 2, 1)$: total probability $\frac{3+5+5}{54} = \frac{13}{54}$. - $W = 2$ when $(1, 2, 2)$: $\frac{8}{54}$. Thus $P(W = 1 \mid X = 1) = \frac{13/54}{21/54} = \frac{13}{21}$, $P(W = 2 \mid X = 1) = \frac{8}{21}$.

   For $X = 2$: - $W = 1$ when $(2, 1, 1)$: $\frac{5}{54}$. - $W = 2$ when $(2, 1, 2)$, $(2, 2, 1)$, $(2, 2, 2)$: $\frac{8+8+12}{54} = \frac{28}{54}$. Thus $P(W = 1 \mid X = 2) = \frac{5/54}{33/54} = \frac{5}{33}$, $P(W = 2 \mid X = 2) = \frac{28}{33}$.

$$\boxed{P(W=1 \mid X=1)=\frac{13}{21},\ P(W=2 \mid X=1)=\frac{8}{21},\ P(W=1 \mid X=2)=\frac{5}{33},\ P(W=2 \mid X=2)=\frac{28}{33}}$$

12. Given the joint pdf

$$f_{X,Y}(x,y)=2e^{-(x+y)}, 0 \le x \le y, y \ge 0$$

find

(a) $P(Y<1|X<1)$
(b) $P(Y<1|X=1)$
(c) $f_{Y|_x}(y)$
(d) $E(Y|_x)$

<u>Sol</u>:

Marginal of X: $f_X(x)=\int_{y=x}^{\infty} 2e^{-(x+y)}dy = 2e^{-x}\int_x^{\infty} e^{-y}dy = 2e^{-2x},\ x \ge 0$.

(a) $P(Y<1 \mid X<1)=\dfrac{P(Y<1, X<1)}{P(X<1)}$.

$P(X<1)=\int_0^1 2e^{-2x}dx = 1-e^{-2}$.

$P(Y<1, X<1) = \int_{x=0}^1 \int_{y=x}^1 2e^{-(x+y)}dy\,dx = \int_0^1 2e^{-x}(e^{-x}-e^{-1})dx = \int_0^1 (2e^{-2x}-2e^{-x-1})dx = [-e^{-2x}+2e^{-x-1}]_0^1 = (-e^{-2}+2e^{-2})-(-1+2e^{-1}) = e^{-2}+1-2e^{-1} = (1-e^{-1})^2$.

Thus $P(Y<1 \mid X<1)=\dfrac{(1-e^{-1})^2}{1-e^{-2}}=\dfrac{1-e^{-1}}{1+e^{-1}}=\dfrac{e-1}{e+1}$.

(b) $P(Y<1 \mid X=1)=0$ because $Y \ge X$ almost surely, and $X=1$ forces $Y \ge 1$.

(c) $f_{Y|X}(y \mid x)=\dfrac{f_{X,Y}(x,y)}{f_X(x)}=\dfrac{2e^{-(x+y)}}{2e^{-2x}}=e^{-(y-x)}, \quad y \ge x$.

(d) $\mathbb{E}[Y \mid X=x]=\int_x^{\infty} y\,e^{-(y-x)}dy$. Substitute $u=y-x$: $=\int_0^{\infty}(x+u)e^{-u}du = x\int_0^{\infty} e^{-u}du + \int_0^{\infty} ue^{-u}du = x\cdot 1+1 = x+1$.

$$\boxed{\text{(a) } \frac{e-1}{e+1} \quad \text{(b) } 0 \quad \text{(c) } e^{-(y-x)},\ y \ge x \quad \text{(d) } x+1}$$

16. Suppose that X and Y are distributed according to the joint pdf

$$f_{X,Y}(x,y)=\frac{2}{5}\cdot(2x+3y), 0 \le x \le 1, 0 \le y \le 1$$

find

(a) $f_X(x)$
(b) $f_{Y|_x}(y)$
(c) $P(\frac{1}{4} \leq Y \leq \frac{3}{4} | X = \frac{1}{2})$
(d) $E(Y|_x)$

<u>Sol</u>:

Given joint pdf $f_{X,Y}(x,y) = \frac{2}{5}(2x+3y)$ for $0 \leq x \leq 1$, $0 \leq y \leq 1$.
(a) Marginal pdf of X:

$$f_X(x) = \int_0^1 \frac{2}{5}(2x+3y)\,dy = \frac{2}{5}\left[2xy + \frac{3}{2}y^2\right]_{y=0}^1 = \frac{2}{5}\left(2x + \frac{3}{2}\right) = \frac{4x+3}{5}, \quad 0 \leq x \leq 1.$$

(b) Conditional pdf of Y given $X = x$:

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{\frac{2}{5}(2x+3y)}{\frac{4x+3}{5}} = \frac{2(2x+3y)}{4x+3} = \frac{4x+6y}{4x+3}, \quad 0 \leq y \leq 1.$$

(c) For $x = \frac{1}{2}$:

$$f_{Y|X}\left(y \,\middle|\, \frac{1}{2}\right) = \frac{4 \cdot \frac{1}{2} + 6y}{4 \cdot \frac{1}{2} + 3} = \frac{2+6y}{5} = \frac{2}{5} + \frac{6}{5}y.$$

Then

$$P\left(\frac{1}{4} \leq Y \leq \frac{3}{4} \mid X = \frac{1}{2}\right) = \int_{1/4}^{3/4} \left(\frac{2}{5} + \frac{6}{5}y\right) dy = \frac{1}{5}\left[2y + 3y^2\right]_{1/4}^{3/4}.$$

At $y = 3/4$: $2 \cdot \frac{3}{4} + 3 \cdot \frac{9}{16} = \frac{3}{2} + \frac{27}{16} = \frac{24+27}{16} = \frac{51}{16}$. At $y = 1/4$: $2 \cdot \frac{1}{4} + 3 \cdot \frac{1}{16} = \frac{1}{2} + \frac{3}{16} = \frac{8+3}{16} = \frac{11}{16}$. Difference: $\frac{51-11}{16} = \frac{40}{16} = \frac{5}{2}$. Then $\frac{1}{5} \cdot \frac{5}{2} = \frac{1}{2}$.
(d) Conditional expectation:

$$\mathbb{E}[Y \mid X = x] = \int_0^1 y \cdot \frac{4x+6y}{4x+3}\,dy = \frac{1}{4x+3}\int_0^1 (4xy + 6y^2)\,dy = \frac{1}{4x+3}\left[2xy^2 + 2y^3\right]_0^1 = \frac{2x+2}{4x+3} = \frac{2(x+1}{4x+3}$$

$$\boxed{f_X(x) = \frac{4x+3}{5}} \quad \boxed{f_{Y|X}(y|x) = \frac{4x+6y}{4x+3}} \quad \boxed{P\left(\frac{1}{4} \leq Y \leq \frac{3}{4} \mid X = \frac{1}{2}\right) = \frac{1}{2}} \quad \boxed{\mathbb{E}[Y \mid X = x] = \frac{2(x+1)}{4x+3}}$$

20. Suppose the random variables X and Y are jointly distributed according to the pdf

$$f_{X,Y}(x,y) = \frac{6}{7}(x^2 + \frac{xy}{2}), 0 \leq x \leq 1, 0 \leq y \leq 2)$$

find

(a) $f_X(x)$

(b) $P(X > 2Y)$

(c) $P(Y > 1 | X > \frac{1}{2})$

<u>Sol</u>:

Given joint pdf $f_{X,Y}(x,y) = \frac{6}{7}\left(x^2 + \frac{xy}{2}\right)$ for $0 \le x \le 1,\ 0 \le y \le 2$.
(a) Marginal $f_X(x)$:

$$f_X(x) = \int_0^2 \frac{6}{7}\left(x^2 + \frac{xy}{2}\right) dy = \frac{6}{7}\left[x^2 y + \frac{x}{2}\cdot\frac{y^2}{2}\right]_0^2 = \frac{6}{7}\left(2x^2 + \frac{x}{2}\cdot\frac{4}{2}\right) = \frac{6}{7}(2x^2+x).$$

(b) $P(X > 2Y) = \int_{x=0}^1 \int_{y=0}^{x/2} \frac{6}{7}\left(x^2 + \frac{xy}{2}\right) dy\, dx$. Inner integral:

$$\int_0^{x/2} \left(x^2 + \frac{xy}{2}\right) dy = x^2 \cdot \frac{x}{2} + \frac{x}{2}\cdot\frac{(x/2)^2}{2} = \frac{x^3}{2} + \frac{x}{2}\cdot\frac{x^2}{8} = \frac{x^3}{2} + \frac{x^3}{16} = \frac{9x^3}{16}.$$

Multiply by $\frac{6}{7}$: $\frac{6}{7}\cdot\frac{9}{16}x^3 = \frac{54}{112}x^3 = \frac{27}{56}x^3$. Then

$$P = \frac{27}{56}\int_0^1 x^3 dx = \frac{27}{56}\cdot\frac{1}{4} = \frac{27}{224}.$$

(c) $P(Y > 1 \mid X > \frac{1}{2}) = \frac{P(Y>1,\, X>\frac{1}{2})}{P(X>\frac{1}{2})}$.

First, $P(X > \frac{1}{2}) = \int_{1/2}^1 f_X(x)dx = \int_{1/2}^1 \frac{6}{7}(2x^2+x)dx = \frac{6}{7}\left[\frac{2}{3}x^3 + \frac{1}{2}x^2\right]_{1/2}^1$. At $x = 1$: $\frac{2}{3} + \frac{1}{2} = \frac{7}{6}$. At $x = \frac{1}{2}$: $\frac{2}{3}\cdot\frac{1}{8} + \frac{1}{2}\cdot\frac{1}{4} = \frac{1}{12} + \frac{1}{8} = \frac{5}{24}$. Difference: $\frac{7}{6} - \frac{5}{24} = \frac{28}{24} - \frac{5}{24} = \frac{23}{24}$. Then $P(X > \frac{1}{2}) = \frac{6}{7}\cdot\frac{23}{24} = \frac{23}{28}$.
Numerator: $P(Y > 1, X > \frac{1}{2}) = \int_{x=1/2}^1 \int_{y=1}^2 \frac{6}{7}\left(x^2 + \frac{xy}{2}\right) dy\, dx$. Inner integral:

$$\int_1^2 \left(x^2 + \frac{xy}{2}\right) dy = x^2(2-1) + \frac{x}{2}\cdot\frac{2^2-1^2}{2} = x^2 + \frac{x}{2}\cdot\frac{3}{2} = x^2 + \frac{3x}{4}.$$

Then $\frac{6}{7}\int_{1/2}^1 \left(x^2 + \frac{3x}{4}\right) dx = \frac{6}{7}\left[\frac{x^3}{3} + \frac{3}{4}\cdot\frac{x^2}{2}\right]_{1/2}^1 = \frac{6}{7}\left[\frac{x^3}{3} + \frac{3x^2}{8}\right]_{1/2}^1$. At $x = 1$: $\frac{1}{3} + \frac{3}{8} = \frac{8}{24} + \frac{9}{24} = \frac{17}{24}$. At $x = \frac{1}{2}$: $\frac{1}{3}\cdot\frac{1}{8} + \frac{3}{8}\cdot\frac{1}{4} = \frac{1}{24} + \frac{3}{32} = \frac{4}{96} + \frac{9}{96} = \frac{13}{96}$. Difference: $\frac{17}{24} - \frac{13}{96} = \frac{68}{96} - \frac{13}{96} = \frac{55}{96}$. Multiply: $\frac{6}{7}\cdot\frac{55}{96} = \frac{330}{672} = \frac{55}{112}$. Thus $P(Y > 1 \mid X > \frac{1}{2}) = \frac{55/112}{23/28} = \frac{55}{112}\cdot\frac{28}{23} = \frac{55}{4\cdot 23} = \frac{55}{92}$.

$\boxed{f_X(x) = \frac{6}{7}(2x^2+x)}$ $\boxed{P(X > 2Y) = \frac{27}{224}}$ $\boxed{P(Y > 1 \mid X > \frac{1}{2}) = \frac{55}{92}}$

Homework 11 - 3.12

Due May 9
Uzair Hamed Mohammed

3.12 - Moment-Generating Functions

4. Find the moment-generating function for the discrete random variable X whose probability function is given by

$$p_X(k) = (\frac{3}{4})^k(\frac{1}{4}), k = 0, 1, 2, \ldots$$

<u>Sol</u>:

The probability function is that of a geometric distribution with success probability $p = \frac{1}{4}$ counting failures:

$$p_X(k) = \left(\frac{3}{4}\right)^k \frac{1}{4}, \quad k = 0, 1, 2, \ldots$$

The moment-generating function is

$$M_X(t) = \mathrm{E}[e^{tX}] = \sum_{k=0}^{\infty} e^{tk} \left(\frac{3}{4}\right)^k \frac{1}{4} = \frac{1}{4} \sum_{k=0}^{\infty} \left(\frac{3e^t}{4}\right)^k.$$

The series converges for $\frac{3e^t}{4} < 1$, i.e., $t < \ln \frac{4}{3}$. Summing gives

$$M_X(t) = \frac{1}{4} \cdot \frac{1}{1 - \frac{3}{4}e^t} = \frac{1}{4 - 3e^t}, \qquad t < \ln \frac{4}{3}.$$

8. Let Y be a continuous random variable with $f_Y(y) = ye^{-y}, 0 \leq y$. Show that $M_Y(t) = \frac{1}{(1-t)^2}$.

<u>Sol</u>:

$$M_Y(t) = \mathrm{E}[e^{tY}] = \int_0^{\infty} e^{ty} y e^{-y}\, dy = \int_0^{\infty} y e^{-(1-t)y}\, dy, \quad t < 1.$$

Let $u = (1-t)y$, $du = (1-t)dy$, $y = u/(1-t)$:

$$M_Y(t) = \int_0^{\infty} \frac{u}{1-t} e^{-u} \frac{du}{1-t} = \frac{1}{(1-t)^2} \int_0^{\infty} u e^{-u}\, du = \frac{\Gamma(2)}{(1-t)^2} = \frac{1}{(1-t)^2}.$$

14. Find an expression for $E(Y^k)$ if $M_Y(t) = (1 - t/\lambda)^{-\lambda}$, where λ is any positive real number and r is a positive integer.

Sol:

$$M_Y(t) = \left(1 - \frac{t}{\lambda}\right)^{-\lambda} \quad \Rightarrow \quad Y \sim \text{Gamma}(\alpha = \lambda,\ \beta = \lambda).$$

$$\mathrm{E}[Y^k] = \frac{\lambda^\lambda}{\Gamma(\lambda)} \int_0^\infty y^{\lambda+k-1} e^{-\lambda y}\, dy = \frac{\lambda^\lambda}{\Gamma(\lambda)} \cdot \frac{\Gamma(\lambda+k)}{\lambda^{\lambda+k}} = \frac{\Gamma(\lambda+k)}{\Gamma(\lambda)\,\lambda^k}.$$

Thus

$$\boxed{\mathrm{E}[Y^k] = \frac{(\lambda)_k}{\lambda^k} = \prod_{i=0}^{k-1}\left(1 + \frac{i}{\lambda}\right)}.$$

16. Find the variance Y is $M_Y(t) = e^{2t}/(1-t^2)$.

Sol:

$$M_Y(t) = \frac{e^{2t}}{1-t^2}, \quad |t| < 1.$$

$$M_Y(t) = \left(1 + 2t + \frac{(2t)^2}{2!} + \frac{(2t)^3}{3!} + \cdots\right)\left(1 + t^2 + t^4 + \cdots\right).$$

Coefficient of t: $2 \Rightarrow E[Y] = 2$.

Coefficient of t^2: from $\frac{(2t)^2}{2!} = 2t^2$ and $1 \cdot t^2 = t^2$ gives $3t^2 \Rightarrow \frac{E[Y^2]}{2!} = 3 \Rightarrow E[Y^2] = 6$.

$$\mathrm{Var}(Y) = 6 - 2^2 = 2.$$

20. Calculate $P(X \le 2)$ if $M_X(t) = (\frac{1}{4} + \frac{3}{4}e^t)^5$.

Sol:

$$M_X(t) = \left(\frac{1}{4} + \frac{3}{4}e^t\right)^5 \implies X \sim \text{Binomial}(n = 5, p = \tfrac{3}{4}).$$

$$P(X \le 2) = \sum_{k=0}^{2} \binom{5}{k} \left(\frac{3}{4}\right)^k \left(\frac{1}{4}\right)^{5-k} = \frac{1}{4^5} \sum_{k=0}^{2} \binom{5}{k} 3^k.$$

$$\binom{5}{0}3^0 = 1, \quad \binom{5}{1}3^1 = 15, \quad \binom{5}{2}3^2 = 10 \cdot 9 = 90.$$

Sum $= 1 + 15 + 90 = 106$. $4^5 = 1024$.

$$P(X \le 2) = \frac{106}{1024} = \frac{53}{512}.$$

22. Suppose the moment-generating function for a random variable W is given by

$$M_W(t) = e^{-3+3e^t} \cdot (\frac{2}{3} + \frac{1}{3}e^t)^4$$

Calculate $P(W \leq 1)$. (Hint: Write W as a sum.)

<u>Sol</u>:

$$M_W(t) = e^{-3+3e^t} \left(\frac{2}{3} + \frac{1}{3}e^t\right)^4$$

implies $W = X + Y$ with $X \sim$ Poisson(3), $Y \sim$ Binomial(4, $\frac{1}{3}$), independent.

$$P(W = 0) = P(X = 0)P(Y = 0) = e^{-3}\left(\frac{2}{3}\right)^4 = e^{-3} \cdot \frac{16}{81}.$$

$$P(W = 1) = P(X = 1)P(Y = 0) + P(X = 0)P(Y = 1)$$

$$= 3e^{-3} \cdot \frac{16}{81} + e^{-3} \cdot \binom{4}{1}\frac{1}{3}\left(\frac{2}{3}\right)^3 = \frac{48}{81}e^{-3} + e^{-3} \cdot \frac{32}{81} = \frac{80}{81}e^{-3}.$$

$$P(W \leq 1) = \left(\frac{16}{81} + \frac{80}{81}\right)e^{-3} = \frac{96}{81}e^{-3} = \frac{32}{27}e^{-3}.$$

24. Suppose that Y is a normal variable, where $f_Y(y) = (1/\sqrt{2\pi}\sigma)\ \exp\ [-\frac{1}{2}(\frac{y-\mu}{\sigma})^2], -\infty < y < \infty$.

(a) Does the random variable $W = 3Y$ have a normal distribution?

(b) DOes the random variable $W = 3Y + 1$ have a normal distribution?

<u>Sol</u>:

$$Y \sim N(\mu, \sigma^2) \implies aY + b \sim N(a\mu + b, a^2\sigma^2).$$

(a) $W = 3Y$: $a = 3, b = 0 \Rightarrow$ normal.

(b) $W = 3Y + 1$: $a = 3, b = 1 \Rightarrow$ normal.

Homework 12 - 4.2, 4.3

Due May 16
Uzair Hamed Mohammed

4.2 The Poisson Distribution

4. A chromosome mutation linked with colorblindness is known to occur, on the average, once in every ten thousand births.

   (a) Approximate the probability that exactly three of the next twenty thousand babies born will have the mutation.

   (b) How many babies out of the next twenty thousand would have to be born with the mutation to convince you that the "one in ten thousand" estimate is too low?

   Sol:

   Let $p = 1/10000$, $n = 20000$, so $\lambda = np = 2$.

   (a) $P(X = 3) \approx e^{-2}\frac{2^3}{3!} = \frac{4}{3}e^{-2} \approx 0.1804$.

   (b) Find smallest k with $P(X \geq k \mid \lambda = 2) \leq 0.05$: $P(X \geq 5) = 0.0527 > 0.05$, $P(X \geq 6) = 0.0166 < 0.05 \Rightarrow k = 6$.

$$\boxed{\text{(a) } \frac{4}{3}e^{-2} \approx 0.1804 \qquad \text{(b) } 6}$$

6. A newly formed life insurance company has underwritten term policies on 120 women between the ages of forty and forty-four. Suppose that each woman has a 1/150 probability of dying during the next calendar year, and that each death requires the company to pay out \$50,000 in benefits. Approximate the probability that the company will have to pay at least \$150,000 in benefits next year.

   Sol:

   Let X be number of deaths. $X \sim \text{Binomial}(n = 120, p = 1/150)$. $\lambda = np = 120/150 = 0.8$. Required probability: $P(X \geq 3) = 1 - P(X = 0) - P(X = 1) - P(X = 2)$. Poisson approximation:

$$P(X = 0) = e^{-0.8} \approx 0.449329,$$
$$P(X = 1) = 0.8e^{-0.8} \approx 0.359463,$$
$$P(X = 2) = \frac{0.8^2}{2}e^{-0.8} = 0.32e^{-0.8} \approx 0.143785.$$

   Sum ≈ 0.952577, so $P(X \geq 3) \approx 0.047423$.

$$\boxed{0.0474}$$

18. Assume that the number of hits, X, that a baseball team makes in a nine-inning game has a Poisson distribution. If the probability that a team makes zero hits is $\frac{1}{3}$, what are their chances of getting two or more hits?

<u>Sol</u>:

Let $X \sim \text{Poisson}(\lambda)$. Given $P(X = 0) = \frac{1}{3}$. Then $e^{-\lambda} = \frac{1}{3}$ implies $\lambda = \ln 3$.

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - \frac{1}{3} - \lambda e^{-\lambda} = 1 - \frac{1}{3} - \frac{\ln 3}{3} = \frac{2 - \ln 3}{3}.$$

$$\boxed{\frac{2 - \ln 3}{3}}$$

24. Let X and Y be independent Poisson random variables with parameters λ and μ respectively. Example 3.12.10 established that X + Y is also Poisson with parameter $\lambda + \mu$. Prove that same result using Theorem 3.8.3.

<u>Sol</u>:

Let $M_X(t) = e^{\lambda(e^t - 1)}$ and $M_Y(t) = e^{\mu(e^t - 1)}$ be the moment generating functions of X and Y. By independence and Theorem 3.8.3,

$$M_{X+Y}(t) = M_X(t) M_Y(t) = e^{\lambda(e^t - 1)} e^{\mu(e^t - 1)} = e^{(\lambda + \mu)(e^t - 1)},$$

which is the mgf of a Poisson random variable with parameter $\lambda + \mu$. Uniqueness of mgf implies $X + Y \sim \text{Poisson}(\lambda + \mu)$.

$$\boxed{X + Y \sim \text{Poisson}(\lambda + \mu)}$$

28. Fifty spotlights have just been installed in an outdoor security system. According to the manufacturer's specifications, these particular lights are expected to burn out at the rate of 1.1 per one hundred hours. What is the expected number of bulbs that will fail to last for at least seventy-five hours?

<u>Sol</u>:

The burn-out rate is 1.1 per 100 hours, so the hazard rate $\lambda = \frac{1.1}{100} = 0.011$ per hour. Lifetime $T \sim \text{Exponential}(\lambda)$. Probability a bulb fails within 75 hours:

$$P(T \leq 75) = 1 - e^{-\lambda \cdot 75} = 1 - e^{-0.011 \times 75} = 1 - e^{-0.825}.$$

Expected number of failures among 50 bulbs:

$$50\left(1 - e^{-0.825}\right) \approx 50 \times 0.5617 = 28.085.$$

$$\boxed{50\left(1 - e^{-0.825}\right) \approx 28.1}$$

4.3 The Normal Distribution

10. State Tech's basketball team, the Fighting Logarithms, have a 70% foul-shooting percentage.

    (a) Write a formula for the exact probability that out of their next one hundered free throws, they will make between seventy-five and eighty, inclusive.
    Sol:

$$P(75 \leq X \leq 80) = \sum_{k=75}^{80} \binom{100}{k} (0.7)^k (0.3)^{100-k}.$$

    (b) Approximate the probability asked for in the above part.
    Sol:

$$\begin{aligned} P(75 \leq X \leq 80) &\approx \Phi\left(\frac{80.5-70}{\sqrt{21}}\right) - \Phi\left(\frac{74.5-70}{\sqrt{21}}\right) \\ &= \Phi(2.2913) - \Phi(0.9820) \\ &= \approx 0.9890 - 0.8365 = \boxed{0.1525}. \end{aligned}$$

18. Suppose $X_1, X_2, X_3,$ and X_4 are independent Poisson random variables each with parameter $\lambda = 3$. Let $S = X_1 + X_2 + X_3 + X_4$

    (a) Use the Central Limit Theorem to approximate the probability that $13 \leq S \leq 14$.
    Sol:

$$\begin{aligned} P(13 \leq S \leq 14) &\approx \Phi\left(\frac{14.5-12}{\sqrt{12}}\right) - \Phi\left(\frac{12.5-12}{\sqrt{12}}\right) \\ &= \Phi(0.7217) - \Phi(0.1443) \\ &\approx 0.7642 - 0.5574 = \boxed{0.2068}. \end{aligned}$$

    (b) Calculate the exact probability that $13 \leq S \leq 14$.
    Sol:

$$P(13 \leq S \leq 14) = e^{-12}\left(\frac{12^{13}}{13!} + \frac{12^{14}}{14!}\right) \approx 0.1056 + 0.0905 = \boxed{0.1961}.$$

26. The cross-sectional area of plastic tubing for use in pulmonary resuscitators is normally distributed with $\mu = 12.5mm^2$ and $\sigma = 0.2mm^2$. When

the area is less than $12.0mm^2$ or greater than $13.0mm^2$, the tube does not fit properly. If the tubes are shipped in boxes of one thousand, how many wrong-sized tubes per box can doctors expect to find?

Sol:

Let $X \sim \mathcal{N}(12.5, 0.2^2)$. Defective if $X < 12$ or $X > 13$.

$$z_1 = \frac{12 - 12.5}{0.2} = -2.5, \quad z_2 = \frac{13 - 12.5}{0.2} = 2.5.$$

$P(\text{defective}) = \Phi(-2.5) + [1 - \Phi(2.5)] = 2\Phi(-2.5) = 2(0.00621) = 0.01242.$

Expected number in box of 1000: $1000 \times 0.01242 = 12.42$.

$$\boxed{12.42}$$

34. Let $Y_1, Y_2, \ldots, Y_n$ be a random sample from a normal distribution where the mean is 2 and the variance is 4. How large must n be in order that

$$P(1.9 \leq \bar{Y} \leq 2.1) \geq 0.99$$

Sol:

Given $\bar{Y} \sim \mathcal{N}\left(2, \frac{4}{n}\right)$. Then

$$P(1.9 \leq \bar{Y} \leq 2.1) = P\left(\frac{1.9 - 2}{2/\sqrt{n}} \leq Z \leq \frac{2.1 - 2}{2/\sqrt{n}}\right) = P\left(|Z| \leq 0.05\sqrt{n}\right).$$

Require $2\Phi(0.05\sqrt{n}) - 1 \geq 0.99 \Rightarrow \Phi(0.05\sqrt{n}) \geq 0.995 \Rightarrow 0.05\sqrt{n} \geq z_{0.005} = 2.576$.

$$\sqrt{n} \geq \frac{2.576}{0.05} = 51.52 \quad \Rightarrow \quad n \geq (51.52)^2 = 2654.79.$$

Thus $\boxed{n = 2655}$.

36. The cylinders and pistons for a certain internal combustion engine are manufactured by a process that gives a normal distribution of cylinder diameters with a mean of 41.5 cm and a standard deviation of 0.4 cm. Similarly, the distribution of piston diameters is normal with a mean of 40.5 cm and a standard deviation of 0.3 cm. If the piston diameter is greater than the cylinder diameter, the former can be reworked until the two "fit." What proportion of cylinder-piston pairs will need to be reworked?

Sol:

Let $X \sim \mathcal{N}(41.5, 0.4^2)$ (cylinder), $Y \sim \mathcal{N}(40.5, 0.3^2)$ (piston), independent. Difference $D = Y - X \sim \mathcal{N}(-1, 0.4^2 + 0.3^2) = \mathcal{N}(-1, 0.25)$. Need $P(Y > X) = P(D > 0) = P\left(Z > \frac{0-(-1)}{0.5}\right) = P(Z > 2) = 1 - \Phi(2) \approx 0.0228$.

$$\boxed{0.0228}$$